

Mechatronics Design Project Report

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1 Introduction

This project involves designing and testing three Simulink modules, including a Pulse Width Modulation and H-bridge-controlled motor, a 10:1 gear train to deliver 100 Nm of torque to the driver joint of Watts 6-bar linkage. The project's aim is to simulate and analyse Watt's six-bar parallel motion linkage in Simulink and Simscape. It is a mechanism that transforms rotational motion into an approximate straight-line motion over a specified range and is commonly used in automotive suspension systems. This report will first present the design for each subsystem and later analyse the force at the linkage output under different spring stiffnesses and damper coefficients of the suspension system to build a functional suspension system that prevents bodies from hitting each other in the 6-bar linkage.

2 DC Motor Circuit

The DC motor Simscape electrical model schematic, which displays the integrated DC Motor, H-Bridge, and Controlled PWM Voltage blocks, is shown in Figure 2.1 below.

2.1 Controlling Speed with PWM

Pulse Width Modulation (PWM) is utilised to govern the rotational speed of the DC motor by varying the average voltage it receives. Rather than supplying a continuous, lowered DC voltage, the PWM signal rapidly switches the full 48V supply on and off in short pulses. The ratio of on time to off time, known as the duty cycle, determines the average voltage. In this system, the duty cycle is set to a constant value below 85%, ensuring the motor receives a proportional average voltage for precise speed control. Furthermore, the PWM block operates at a high frequency of 4000 Hz; this frequency is sufficient for the motor's internal inductance to smooth the rapid current drops, yielding a stable output torque.

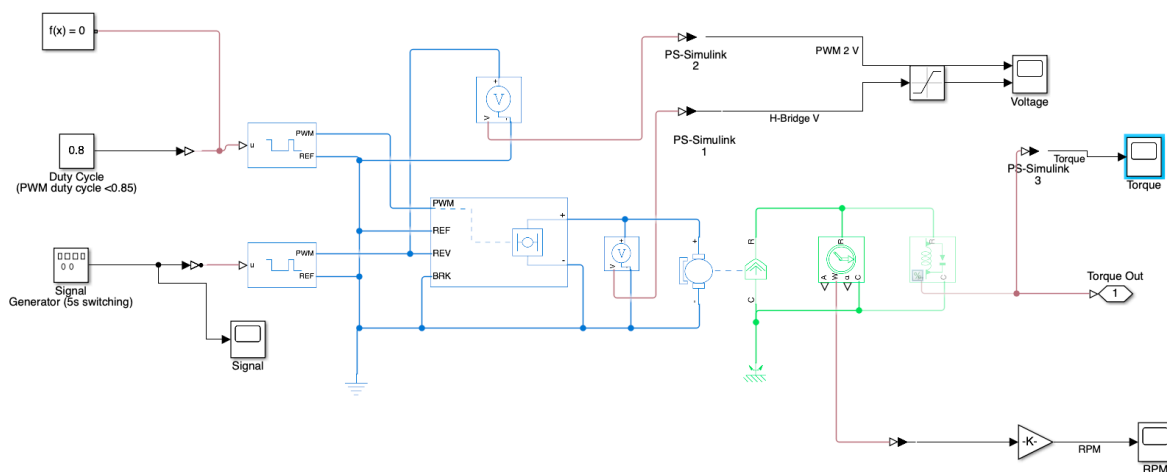


Figure 2.1: Simscape electrical model schematic displaying the integrated DC Motor, H-Bridge, and Controlled PWM Voltage blocks.

2.2 Controlling Direction with the H-Bridge

While the PWM controls the motor speed, the H-Bridge controls the motor's direction of rotation. The H-Bridge functions as a set of electronic switches that dictate the direction of current flowing through the motor. This direction is manipulated using the reverse (REV) input port on the H-Bridge. By driving the REV port with a 0.1 Hz square wave with a period of 10 seconds, the incoming voltage crosses the H-Bridge's internal 2.5V threshold exactly every 5 seconds. When the signal is low, the H-Bridge sends current in one direction. When the signal crosses the threshold and goes high, the internal switches flip, reversing the polarity of the 48V supply and causing the motor to spin in the opposite direction. This behaviour is shown in Figure 2.2 below, where the motor speed oscillates between positive and negative operating velocities every 5 seconds. The motor speed and H-bridge amplitude measurements are also expected, with the H-bridge voltage at approximately 38V and the DC motor speed at 1600 rpm (80% of the 48V input and 2000 rpm no-load speed), respectively. The 2000 motor speed and 48V H-bridge motor specifications at 100% duty cycle are also verified and plotted in Appendix D.

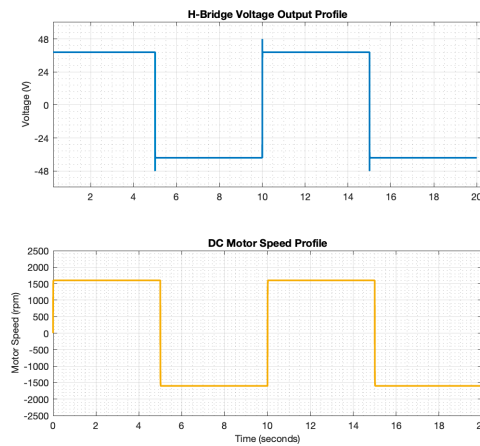


Figure 2.2: H-Bridge terminal voltage and corresponding motor angular velocity profile operating at 80% duty cycle

2.3 Motor Torque

At the chosen 80% duty cycle, simulation results demonstrate that the motor generates a baseline output torque of approximately 10, applied in opposite directions every 5s. This output torque, shown in Figure 2.3 below, serves as the input force for the subsequent gear train system.

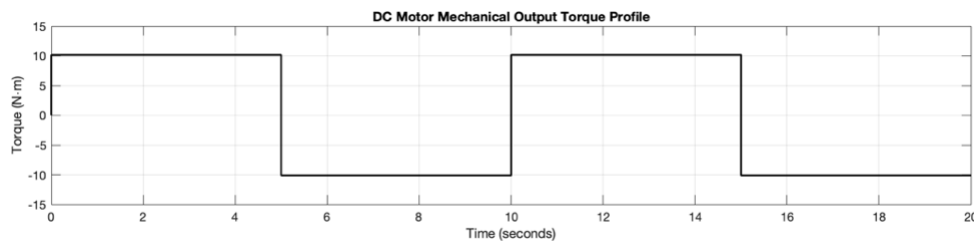


Figure 2.3: DC Motor mechanical output torque profile at an 80% duty cycle

3 Gear Train

Because the DC motor output generates approximately 10 Nm of torque at an 80% duty cycle, a mechanical advantage is required to deliver 100 Nm to Fixed Joint 1 of the linkage to actuate the system under load. Therefore, a 10:1 three-stage gear train is needed to deliver the increased torque while maintaining a gear radius within the specified range of 4 to 10 cm.

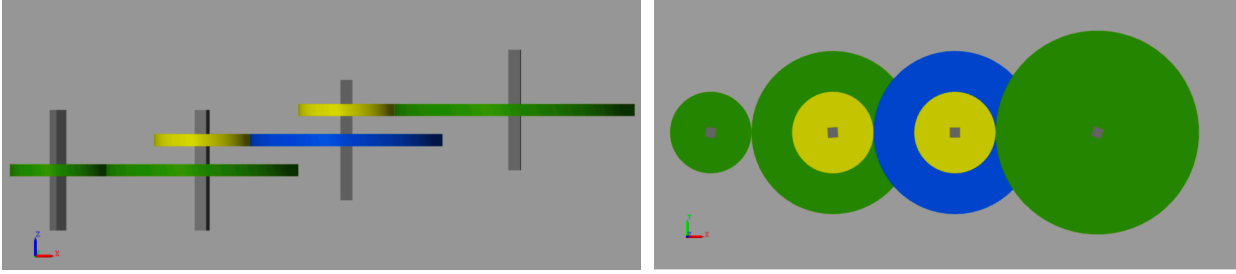


Figure 3.1: Simulink 10:1 gear train design top and side profile

3.1 Gear ratio calculations

The gear ratio for an individual stage is calculated using the ratio of the driven gear radius to the driver gear radius.

$$N = \frac{r_{driven}}{r_{driver}}$$

Stage 1 has a 2:1 gear ratio with driver and driven gears 4cm and 8cm, respectively.

$$N_1 = \frac{8}{4} = 2$$

Stage 2 also has a 2:1 gear ratio with driver and driven gears 4cm and 8cm, respectively.

$$N_2 = \frac{8}{4} = 2$$

Stage 3 implements a 2.5:1 gear ratio with driver and driven gears 4cm and 10cm, respectively.

$$N_3 = \frac{10}{4} = 2.5$$

The total gear ratio of the compound system is the product of the individual stage ratios.

$$N_{total} = N_1 \times N_2 \times N_3 = 2 \times 2 \times 2.5 = 10$$

With a total gear ratio of 10, the output torque τ_{out} is calculated by multiplying the input torque τ_{in} by the gear ratio, assuming ideal and lossless power transmission.

$$\tau_{out} = \tau_{in} \times N_{total} = 10 \times 10 = 100 \text{ Nm}$$

3.2 Simscape Implementation and Verification

The calculated gear dimensions were implemented in the Simscape using common gear constraint blocks between the respective revolute joints and cylindrical solids and is outlined in Figure 3.2 below.

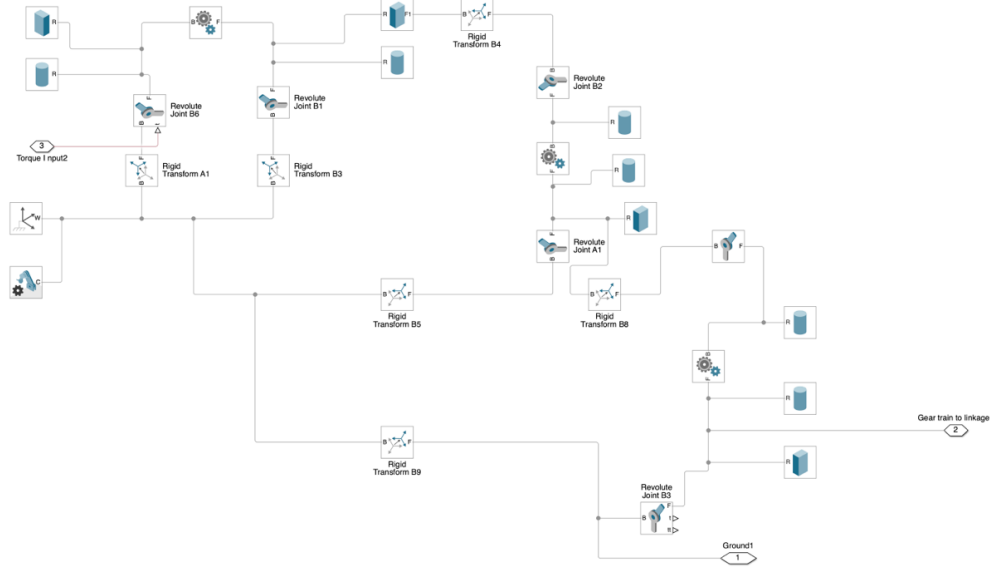


Figure 3.2: Simscape Multibody schematic of the three-stage compound gear train.

Each stage-driven gear is connected to the same axle as the next stage's driver gear to connect subsequent gear stages, as seen in Figure 4. The trade-off between an increase in torque is a decrease in rotational speed. By stepping down the rotational velocity across the three stages, the torque is proportionally multiplied by the gear ratios at each stage. To verify that the gear train provides the desired torque increase, measure the torque at the final revolute joint in Fixed Joint 1 of the six-bar linkage, which is plotted alongside the output motor torque from the first stage.

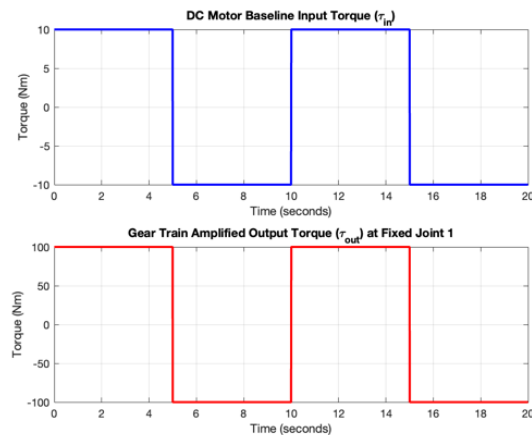


Figure 3.3 Comparison of torque before and after the geartrain.

Figure 3.3 above visually confirms the successful implementation of the design, demonstrating that the motor's initial input force of 10 Nm is accurately amplified to the required 100 Nm target.

4 Watt's 6-Bar Parallel Motion Linkage

The Watt parallel motion linkage is a historic mechanical linkage devised by Scottish engineer James Watt in 1784, and was originally developed to guide the piston rod of early steam engines in a straight cylinder. It was a pivotal invention for the Industrial Revolution. Its primary function is to transform rotational motion into an approximate straight-line motion over a specified range. In automotive suspension systems, it prevents the axle from moving side-to-side relative to the vehicle chassis during cornering, ensuring stability. The Simulink model of this linkage is outlined in Figure 4.1 below.

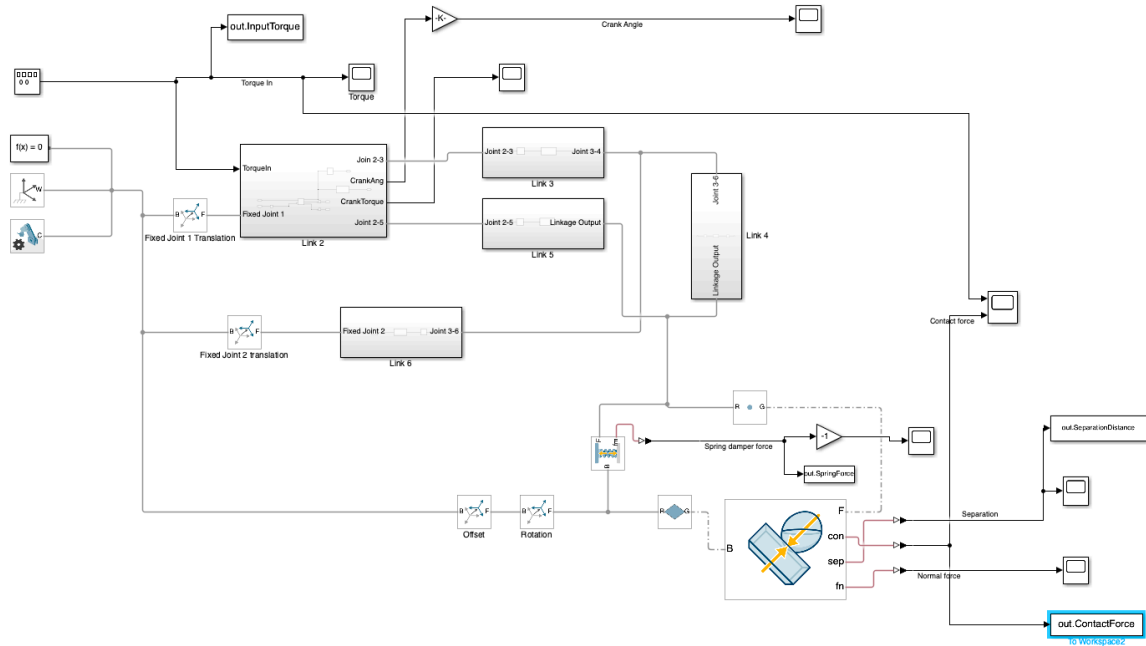


Figure 4.1: Simscape block diagram of 6-bar Watt's linkage.

4.1 Mechanism of Vertical Motion

The specific linkage modelled in this simulation consists of 6 rigid links and 7 joints, as detailed in the schematic in Appendix E. The two fixed revolute joints—fixed joints 1 and two—which allow for rotation but restrict any translational movement. Link 2 serves as the motion driver for the system, receiving torque from the gear train at Fixed Joint 1. The vertical motion is achieved through the geometric constraints imposed by the link lengths. Based on the system parameters, Link 2, which drives the system, has a length of 1 m, Link 3 is 0.4 m, and Links 4 and 6 are 0.5 m each. The horizontal distance between the two fixed joints is 1 m, while their vertical separation is approximately 0.4 m. Link 1 is the ground, which is the rigid 1m horizontal and 0.4m vertical distance between fixed joints 1 and two. Because of the fixed horizontal distance constraint between the revolute anchors and the equal 0.5 m lengths of Links 4 and 6, the rotational input from Link 2 acts against the constraints of the opposing links, which forces the central output link on joint 4-5 to travel along a highly constrained path that is near-perfectly fixed along the y-plane while the join linkage moves up and down as torque is applied to the driving link.

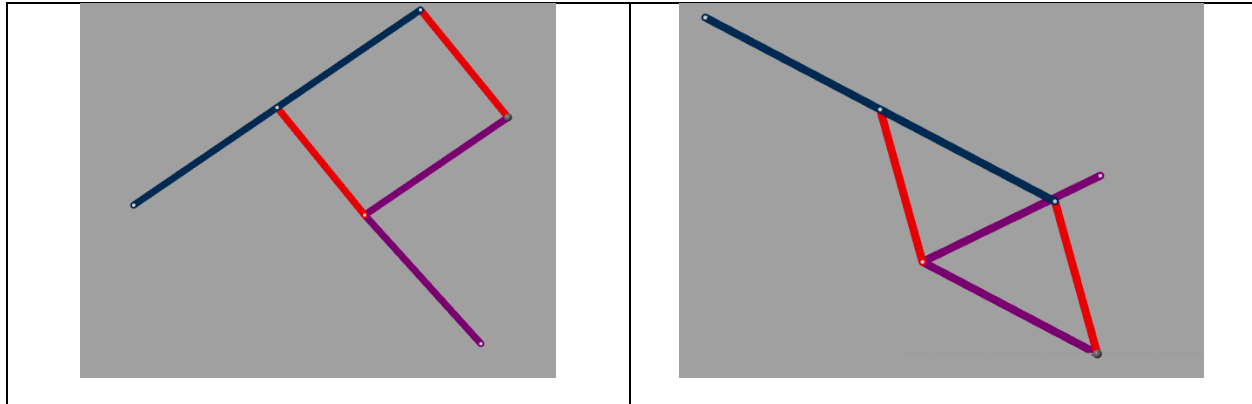


Figure 4.2: Simulink Mechanics Explorer visualisation of the Watt's parallel linkage with maximum vertical extensions

4.2 Maximum Vertical Distance Analysis

While Watt's linkage provides linear vertical motion, it does not provide infinite linear travel. The linkage output travels linearly only for a finite distance before the links' arcs cause the path to change direction. To analyse this, the simulation was executed without the suspension spring attached, allowing the linkage to free-fall towards the ground, which is 0.85m below fixed joint 1. Using the Spatial Contact Force block within Simscape, the separation distance between the linkage output and the infinite plane was tracked over time, and the scope output is shown in Figure 4.3 below. The simulation data indicate that the linkage output achieves a maximum vertical travel distance of 1.14m from the ground, at the maximum point using the scope cursor, before geometric constraints cause the path between the linkage output and the ground to deviate horizontally or the system reaches its mechanical limit when it is fully extended, as shown in Figure 4.2 above. With the fixed joint at 1, 0.85m above the ground as measured by the linkage output (joint 4-5), the mechanism can extend approximately 0.3m above this fixed point until it is restricted by link 6 being fully extended.

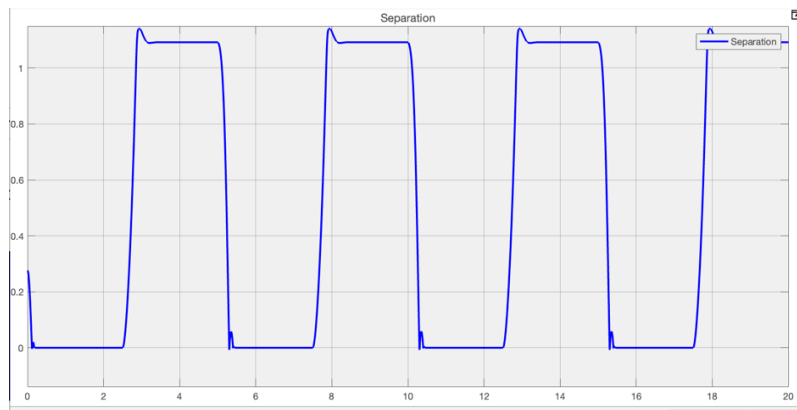


Figure 4.3: Simulink scope plot displaying the Separation Distance vs Time with 100N of input torque applied and no suspension

5 Suspension

The suspension system acts as a mass-spring damper system between the linkage output and the ground. Its primary role is to absorb the force generated by the linkage motion and prevent the output from hitting the ground, thereby reducing high-impact forces. The following case study will evaluate the effects of spring stiffness and damper coefficients.

Spring stiffness, also known as the spring constant, k , is the amount of force required to compress or extend a spring by a specific unit of distance. It measures a spring's resistance to deformation: a higher stiffness means the spring is harder to compress or stretch. The damper coefficient, c , quantifies how much a damper resists motion in a mechanical system and how quickly the system dissipates energy and stops vibrating; its primary role is to absorb and dissipate kinetic energy from a moving system

5.1 Suspension Parameter Case Study

The following parameters were tested in the case study below, and a summary of the experiment parameters is outlined in Table 5.1. The six-bar linkage in the case study is tested independently of the system, and for each study, the Simulink system variables are exported as workspace variables and plotted in MATLAB.

Table 5.1: Case study experiment suspension parameters

Case	Input Torque Magnitude (Nm)	Input Torque Frequency (Hz)	Spring Stiffness (Nm)	Damper Coefficient (N/(m/s))	Spring natural length (m)
1	100	0.2	1000	0	0.3
			750		
			500		
2	100	0.2	1000	20	0.3
				50	
				100	
3	200	0.5	1000	100	0.3
			1100	150	
			1100	200	

Case 1: Analysis of Spring Stiffness with Zero Damping applied

The input torque, force profiles, and separation distance plots are outlined in Appendix A. Under the conditions of case 1, the suspension system exhibits a highly oscillatory, underdamped "bouncy" force profile due to the lack of energy dissipation. It was found that a spring stiffness of $k = 1000$ N/m provides sufficient potential energy storage to prevent the spring from hitting the ground, as the ground contact force is zero in the first case and the suspension force is highest, with the spring damper force around 270 N. Dropping the stiffness to 750 N/m results in insufficient resistance to the linkage's downward momentum, leading to ground contact with a 3 kN normal force. This trend is exacerbated as the spring stiffness decreases to 500 N/m, resulting in 5 kN spikes in ground impact force that could damage the

linkage. Further, in a practical mechatronic application, a suspension operating under these parameters is not ideal; without damping to stabilise oscillations, the repetitive, high-magnitude impact loads would rapidly induce structural fatigue, accelerated component wear, and mechanical failure of the 6-bar linkage joints.

Case 2: Analysis of Damping Coefficients

The input torque, force profiles, and separation distance plots are outlined in Appendix B. In case 2, as the damping coefficients (c) increase, the Spring/Damper force profile has fewer spikes and becomes increasingly damped and settles to a stable value of about 150 N more rapidly. Similar behaviour is observed for the separation distance. At $c=20$ N/(m/s), the system has an underdamped transition response. When the motor switches directions the substantial initial overshoot and oscillatory behaviour. The ringing is further reduced at $c=50$ N/(m/s), and when b is increased to 100 N/(m/s), the oscillations disappear, and the system appears to be critically damped, returning to a constant equilibrium value without oscillating, as shown in Figure 5.1 below. Increasing the damping further could result in overdamping, which increases the time it takes for the system to settle and could make the suspension assembly too rigid against dynamic changes in velocity. Consequently, a damping coefficient of $c=100$ N(m/s) and a spring stiffness of $k=1000$ N/m lead to the optimal compromise, as they provide a sufficient damping ratio to mitigate ringing and overshoots, induce structural fatigue, accelerate components, and prevent the system from hitting the ground.

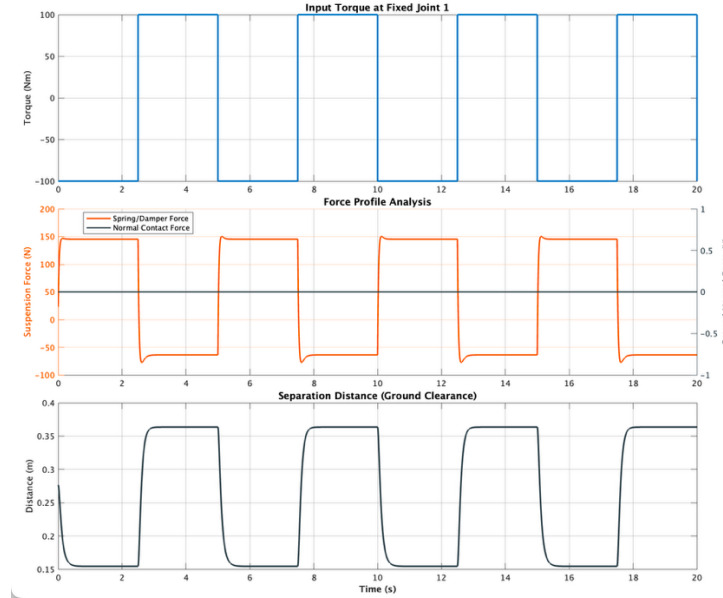


Figure 5.1: Input torque, force profile, and separation distance for case 2 with a spring stiffness of 1000 N/m and a damping coefficient of 100 N/(m/s)

Case 3: Higher Input Torque Magnitude and Frequency Analysis

The input torque, force profiles, and separation distance plots for case 3 are outlined in Appendix C. To reliably increase the torque and frequency in case 3 to 200 Nm and 0.5 Hz, a square wave generator was used and connected directly to the torque input of fixed joint 1 in Simulink. The first experiment involved applying the stiffness-and-damping combination in case 2, where $k = 1000$ N/m and $c = 500$ N/(m/s). It was found that these parameters did not completely reduce the spikes, as the Spring/Damper force profile was slightly underdamped and the spring stiffness was insufficient to absorb incoming forces and prevent bodies from colliding, with normal contact forces spiking at 1.6 kN. To improve the suspension system, the constant k was increased to 1100 N/m, which was sufficient to reduce the force due to ground contact, and the damping coefficient was increased to 150 N/m/s, which reduced spikes in the suspension force, leading to a more stable response. The resulting system Input torque and force profile is outlined in Figure 5.2 below. Further, a more aggressive damping coefficient of 200 N/(m/s) was tested, but it appeared slightly overdamped, increasing the settling time for the separation distance. Therefore, a spring stiffness of $k=1100$ N/m and a damping coefficient of around 150 N/(m/s) led to the most desirable suspension, capable of absorbing incoming forces, preventing bodies from hitting each other, and providing smoother movement, as seen in the separation distance plot.

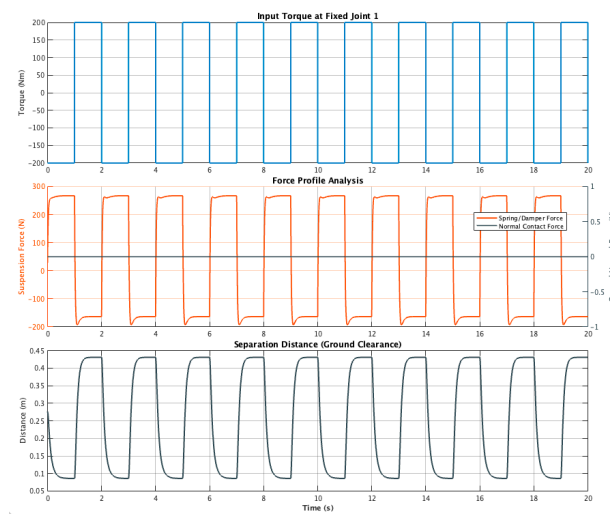


Figure 5.2: Input torque, force profile, and separation distance for case 3 with a spring stiffness of 1100 N/m and a damping coefficient of 150 N/(m/s)

6. Reflection

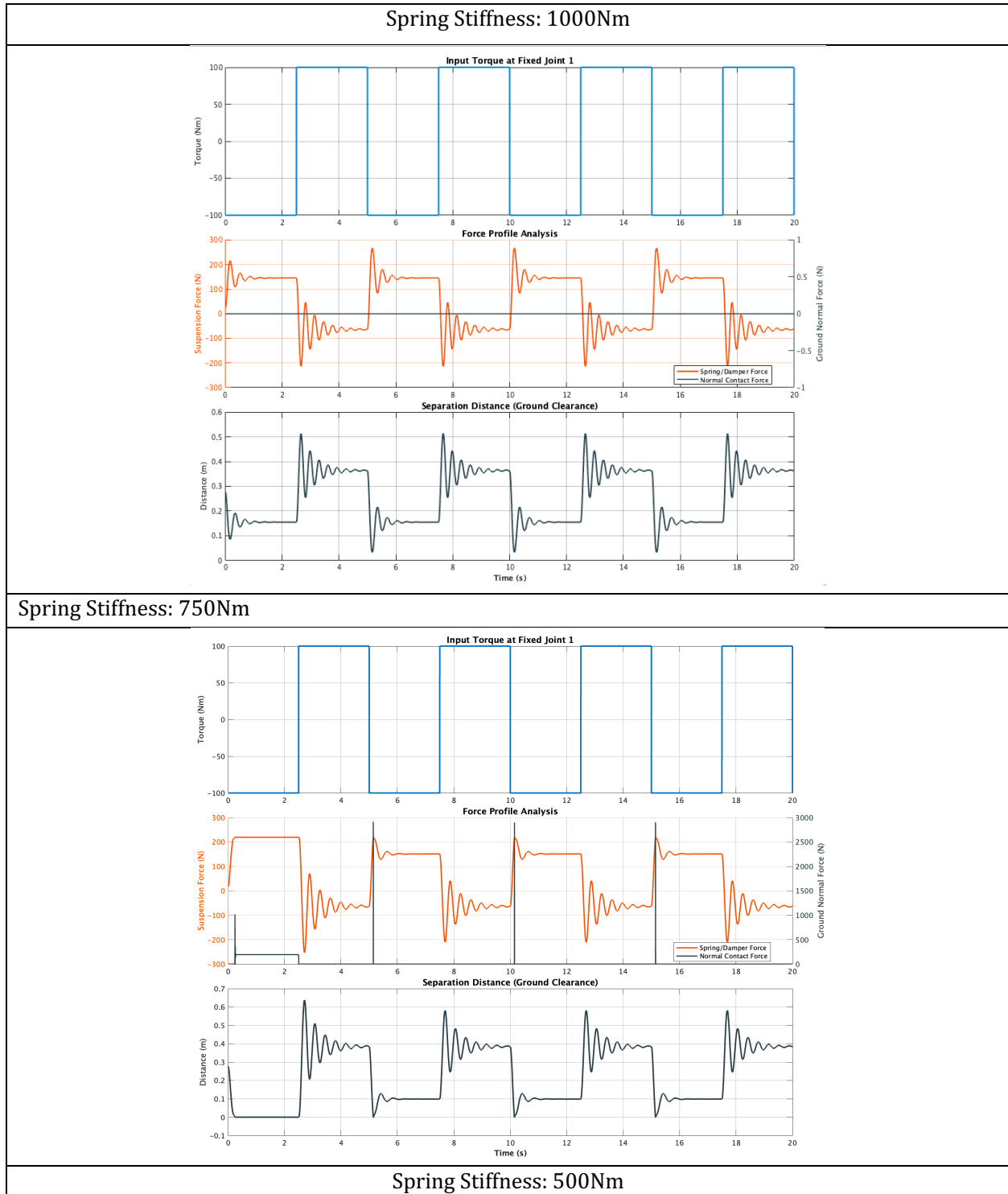
The full system was built incrementally in Simulink by integrating independently tested modules sequentially. The motor subsystem with PWM and an H-bridge was first validated before being connected to the mechanical rotational domain (the 10:1 gear train) to scale up motor torque. Finally, this rotational assembly was integrated with the mechanical translational domain via a revolute joint to drive the full Watt's 6-bar linkage mechanism. Testing each module independently proved crucial to isolating interface mismatches before full-system simulation. Significant engineering challenges were encountered during data analysis and simulation stability. Notably high-frequency imperfections and numerical noise in the torque profile at the fixed gear joint, which diverged slightly from expected values. Furthermore, the analytical ode15s solver produced severe numerical voltage spikes within the PWM H-bridge electrical network. To resolve this and ensure clean data visualisation, a saturation block was successfully implemented to suppress these transient spikes for plotting.

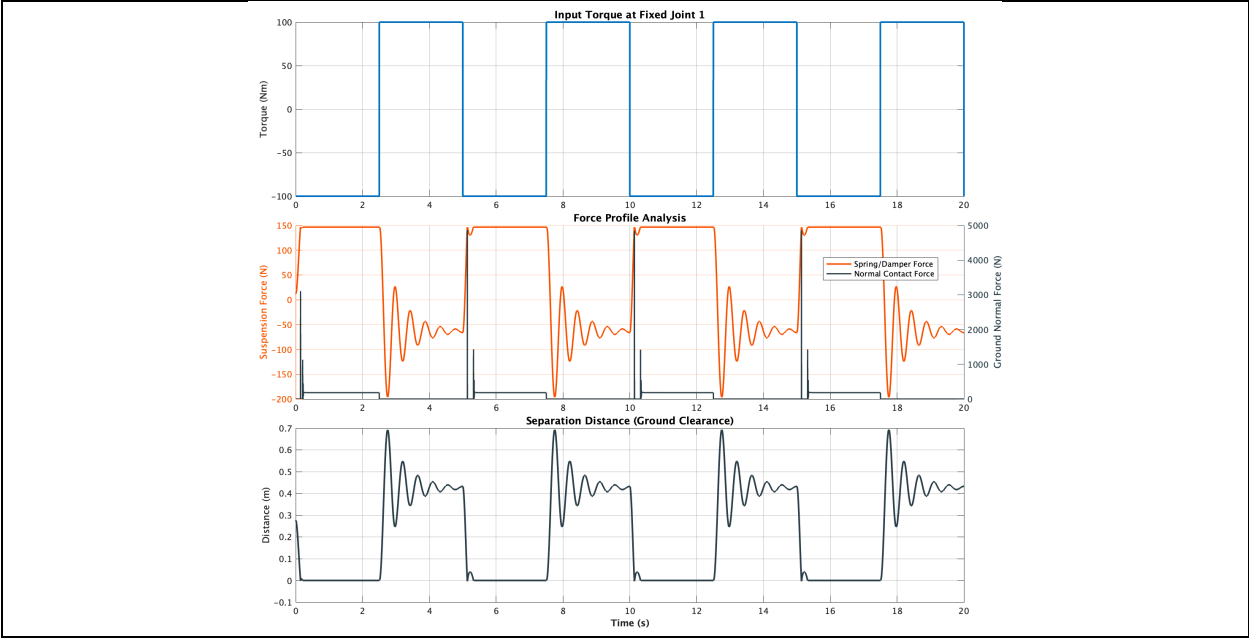
7. Conclusion

The objectives of the design project were achieved through multi-domain Simulink modelling, and the Motor, 10:1 Gear Train, and Watts six-bar linkage were combined and verified to meet specifications, with the full Simulink system schematic outlined in Appendix G. The integrated PWM, H-bridge, and gear train assembly demonstrated a can control motor speed and deliver a 10Nm of torque at 1500rpm to the gear train, switching directions every 5s. The gear train was demonstrated to amplify torque tenfold, delivering 100 Nm to the fixed joint 1 link 2 in Watt's 6-bar linkage. Lastly, as the suspension system connected to the output joint on the 6-bar linkage was analysed, to mitigate unnecessary oscillatory behaviour, preventing any high force ground contact which may damage the suspension. The most appropriate results were found in case study 3 with a spring stiffness of 1100 N/m and a damping coefficient of 150 N/(m/s), which were included in the final model. When Watts' 6-bar linkage is connected to the gear train and motor, it oscillates up and down smoothly, as outlined in Appendix F, serving its intended purpose of transforming rotational motion into an approximate straight-line motion over a specified range.

6 Appendix

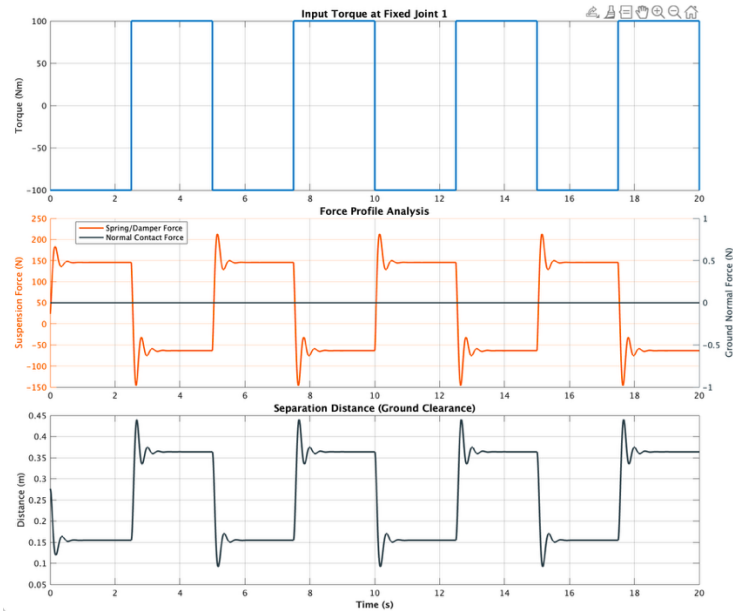
A: Suspension Case Study 1 plots



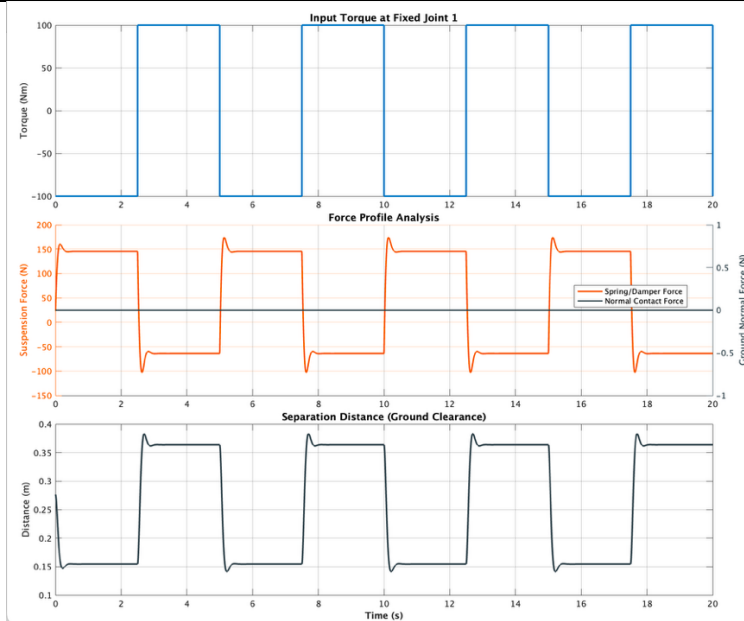


B: Suspension Case Study 2 Plots

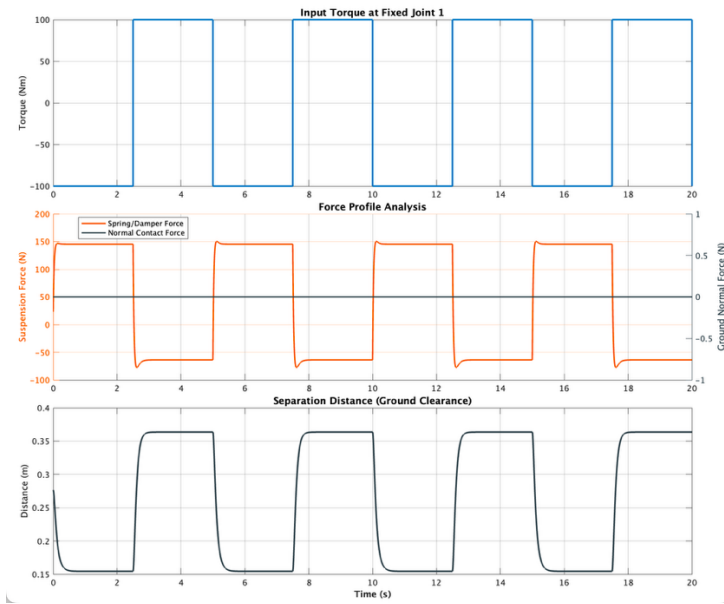
Damper Coefficient: 20 N/(m/s)



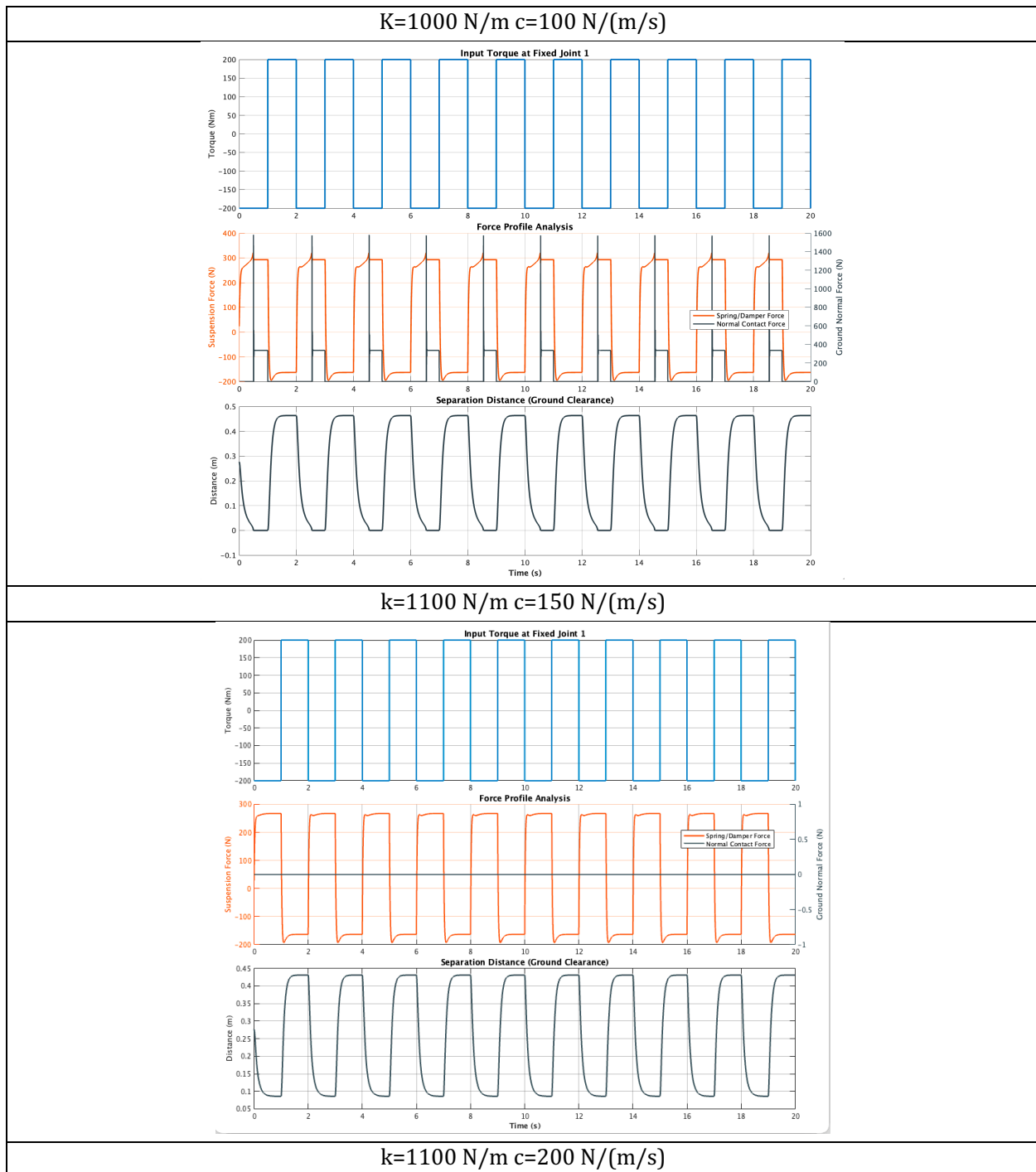
Damper Coefficient: 50 N/(m/s)

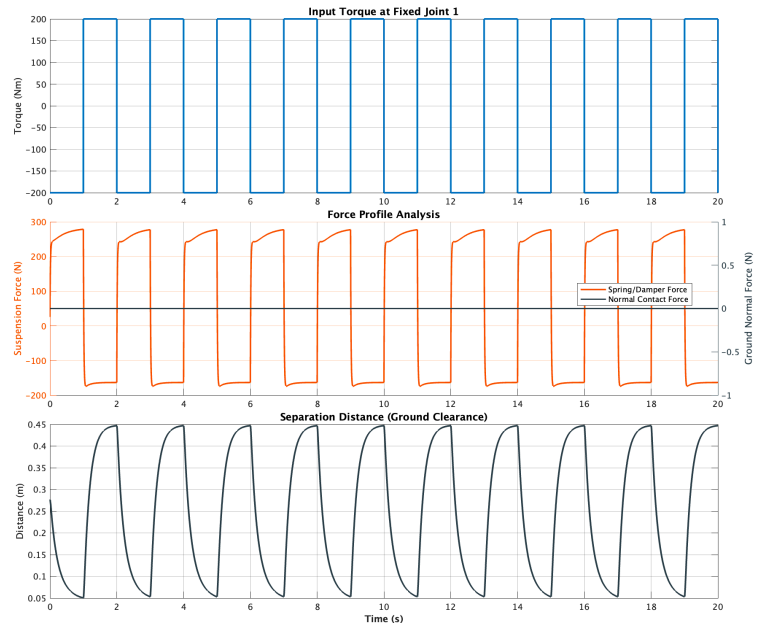


Damper Coefficient: 100 N/(m/s)

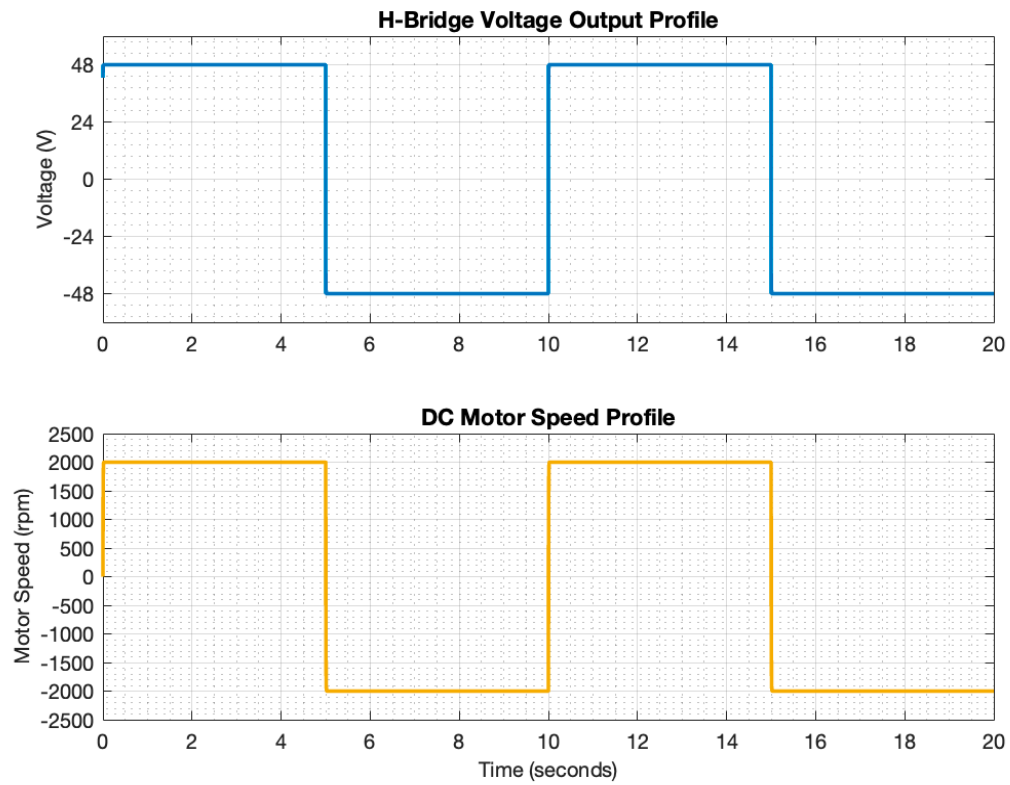


C: Suspension Case Study 3 Plots

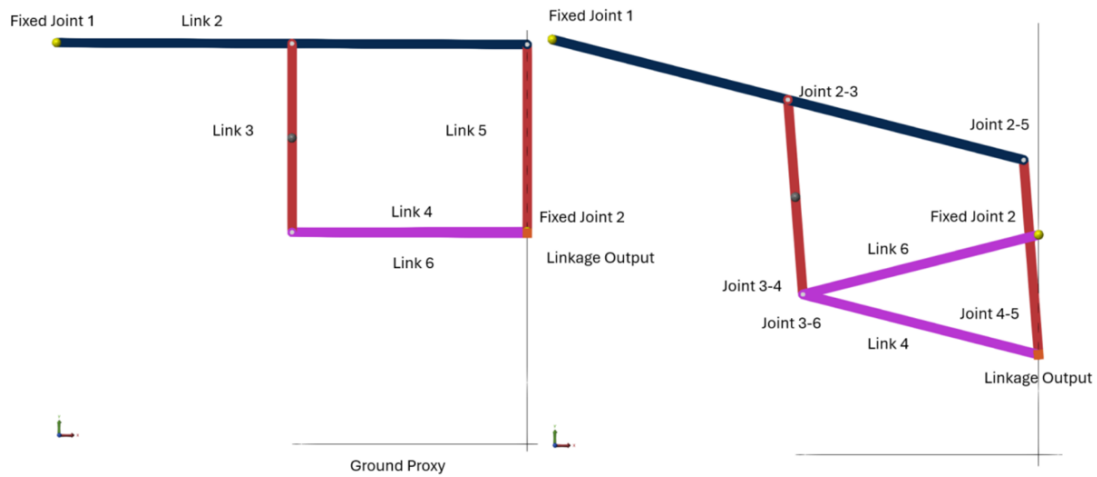




D: H-Bridge motor voltage and speed plot for 100% duty cycle



E: Watts Parallel Linkage Schematic



F: Full System View in Simulink Mechanics Explorer

Parallel
Fully Extended Down

G: Full System Simulink Schematic

